# HOMEWORK 1

Converted from the uploaded document. Contains all 15 programs with source code and sample output.

## 1. Sum of two matrices

Source Code:

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter first matrix:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter second matrix:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            sum[i][j] = a[i][j] + b[i][j];
        }
    }

    printf("Sum of the two matrices:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            printf("%d\t", sum[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter rows and columns: 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8
Sum of the two matrices:
6 8
10 12
```

## 2. Read multiple values into an array and display them in reverse order.

Source Code:

```
#include <stdio.h>

int main()
{
    int arr[100], n, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);

    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Array in reverse order:\n");

    for(i = n - 1; i >= 0; i--)
    {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
Array in reverse order:
50 40 30 20 10
```

## 3. Find the total number of duplicate elements in an array

Source Code:

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, j;
    int count = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(arr[i] == arr[j])
            {
                count++;
                break;
            }
        }
    }

    printf("Total number of duplicate elements = %d", count);

    return 0;
}
```

OUTPUT:

```
Enter the number of elements: 7
Enter the elements:
10 20 10 30 20 40 20
Total number of duplicate elements = 2
```

## 4. Find and print the unique elements in an array

Source Code:

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, j, count;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Unique elements are:\n");

    for(i = 0; i < n; i++)
    {
        count = 0;

        for(j = 0; j < n; j++)
        {
            if(arr[i] == arr[j])
            {
                count++;
            }
        }

        if(count == 1)
        {
            printf("%d ", arr[i]);
        }
    }

    return 0;
}
```

OUTPUT:

```
Enter the number of elements: 7
Enter the elements:
10 20 10 30 20 40 50
Unique elements are:
30 40 50
```

## 5. Insert an element into a sorted array while keeping the array sorted

Source Code:

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, val;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter elements in sorted order:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter the value to insert: ");
    scanf("%d", &val);

    for(i = n - 1; i >= 0 && arr[i] > val; i--)
    {
        arr[i + 1] = arr[i];
    }

    arr[i + 1] = val;
    n++;

    printf("Array after insertion:\n");

    for(i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter the number of elements: 5
Enter elements in sorted order:
10 20 30 40 50
Enter the value to insert: 35
Array after insertion:
10 20 30 35 40 50
```

## 6. Insert an element at a given position in an unsorted array

Source Code:

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, pos, val;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter the value to insert: ");
    scanf("%d", &val);

    printf("Enter the position: ");
    scanf("%d", &pos);

    if(pos < 1 || pos > n + 1)
    {
        printf("Invalid position");
        return 0;
    }

    for(i = n; i >= pos; i--)
    {
        arr[i] = arr[i - 1];
    }

    arr[pos - 1] = val;
    n++;

    printf("Array after insertion:\n");

    for(i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter the number of elements: 5
Enter the elements:
10 50 20 40 30
Enter the value to insert: 25
Enter the position: 3
Array after insertion:
10 50 25 20 40 30
```

## 7. Matrix multiplication

Source Code:

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], product[10][10];
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter rows and columns of second matrix: ");
    scanf("%d %d", &r2, &c2);

    if(c1 != r2)
    {
        printf("Matrix multiplication is not possible.");
        return 0;
    }

    printf("Enter first matrix:\n");

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c1; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter second matrix:\n");

    for(i = 0; i < r2; i++)
    {
        for(j = 0; j < c2; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            product[i][j] = 0;

            for(k = 0; k < c1; k++)
            {
                product[i][j] += a[i][k] * b[k][j];
            }
        }
    }

    printf("Product of matrices:\n");

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            printf("%d\t", product[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
```

```
5 6
7 8
Product of matrices:
19 22
43 50
```

## 8. Find the transpose of a matrix

Source Code:

```
#include <stdio.h>

int main()
{
    int a[10][10], t[10][10];
    int r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter matrix elements:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            t[j][i] = a[i][j];
        }
    }

    printf("Transpose of the matrix:\n");

    for(i = 0; i < c; i++)
    {
        for(j = 0; j < r; j++)
        {
            printf("%d\t", t[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter rows and columns: 2 3
Enter matrix elements:
1 2 3
4 5 6
Transpose of the matrix:
1 4
2 5
3 6
```

## 9. Right diagonal sum of a square matrix

Source Code:

```
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, sum = 0;

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < n; i++)
    {
        sum = sum + a[i][n - 1 - i];
    }

    printf("Right diagonal sum = %d", sum);

    return 0;
}
```

OUTPUT:

```
Enter the size of matrix: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9
Right diagonal sum = 15
```

## 10. Check for identity matrix

Source Code:

```
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, flag = 1;

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i == j && a[i][j] != 1)
            {
                flag = 0;
            }
            else if(i != j && a[i][j] != 0)
            {
                flag = 0;
            }
        }
    }

    if(flag == 1)
        printf("The matrix is an identity matrix.");
    else
        printf("The matrix is not an identity matrix.");

    return 0;
}
```

OUTPUT:

```
Enter the size of matrix: 3
Enter matrix elements:
1 0 0
0 1 0
0 0 1
The matrix is an identity matrix.
```

## 11. Count the number of vowels, consonants, digits, and white spaces in a string

Source Code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((str[i] >= 'a' && str[i] <= 'z') ||
           (str[i] >= 'A' && str[i] <= 'Z'))
        {
            if(str[i] == 'a' || str[i] == 'e' ||
               str[i] == 'i' || str[i] == 'o' ||
               str[i] == 'u' || str[i] == 'A' ||
               str[i] == 'E' || str[i] == 'I' ||
               str[i] == 'O' || str[i] == 'U')
            {
                vowels++;
            }
            else
            {
                consonants++;
            }
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if(str[i] == ' ' || str[i] == '\t')
        {
            spaces++;
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
    printf("Digits = %d\n", digits);
    printf("White spaces = %d\n", spaces);

    return 0;
}
```

OUTPUT:

```
Enter a string: Hello World 123
Vowels = 3
Consonants = 7
Digits = 3
White spaces = 2
```

## 12. Find the frequency of each character in a string

Source Code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int freq[256] = {0};
    int i;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        freq[(unsigned char)str[i]]++;
    }

    printf("Character frequencies:\n");

    for(i = 0; i < 256; i++)
    {
        if(freq[i] > 0 && i != '\n')
        {
            printf("%c = %d\n", i, freq[i]);
        }
    }

    return 0;
}
```

OUTPUT:

```
Enter a string: hello
Character frequencies:
e = 1
h = 1
l = 2
o = 1
```

## 13. Count the number of words in a string

Source Code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, words = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((i == 0 && str[i] != ' ' && str[i] != '\n') ||
           (str[i] != ' ' && str[i] != '\n' && str[i - 1] == ' '))
        {
            words++;
        }
    }

    printf("Number of words = %d", words);

    return 0;
}
```

OUTPUT:

```
Enter a string: Hello world this is C
Number of words = 5
```

## 14. Sort an array of strings in alphabetical order

Source Code:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[10][100], temp[100];
    int n, i, j;

    printf("Enter number of strings: ");
    scanf("%d", &n);

    printf("Enter %d strings:\n", n);

    for(i = 0; i < n; i++)
    {
        scanf("%s", str[i]);
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Strings in alphabetical order:\n");

    for(i = 0; i < n; i++)
    {
        printf("%s\n", str[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter number of strings: 5
Enter 5 strings:
banana
apple
mango
grapes
orange
Strings in alphabetical order:
apple
banana
grapes
mango
orange
```

## 15. Toggle the case of every alphabet in a sentence

Source Code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
        else if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
    }

    printf("Toggled sentence: %s", str);

    return 0;
}
```

OUTPUT:

```
Enter a sentence: Hello World! This Is C Programming.
Toggled sentence: hELLO wORLD! tHIS iS c pROGRAMMING.
```